

Self-Fulfilling Prophecy?: Predicting Longevity and Mortality Using Subjective Mortality Expectations in the HRS

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1 Background

The extent to which we can predict mortality well matters for understanding population health. Recent work shows that individual-level mortality and longevity prediction is challenging, with uncertainty remaining high even with big microdata and rich individual-level covariates ([Badolato et al. 2023](#); [Breen and Seltzer 2023](#)). Past work has employed subjective mortality expectations to aid in interpreting the predictive performance of models, but overlooked their role as focal features. While individuals are not very good at predicting their own mortality and there are noted inequalities in accuracy by social group, whether subjective probabilities, however flawed, can be used in a prediction model as “utility features” ([Kose et al. n.d.](#)) in combination with other strongly predictive variables (e.g., education, income, age, health) to achieve gains in predictive performance is currently unexplored. This project builds upon past efforts by testing the predictive performance of subjective life expectancy measures to (1) predict individual longevity and (2) predict survival past threshold ages (e.g., 75 years) relative to models without expectations.

2 Policy Relevance

There are a variety of substantive and policy relevant reasons for trying to obtain accurate mortality predictions. First, predicting mortality from individual-level data is important for forecasting population longevity in the future, especially by social groups or categories that are not readily available from tabular data. Generating accurate predictions can inform public health policy decisions allocating resources to populations at older ages based on predicted life expectancy with morbidities, disabilities, disease, loneliness, and other chronic conditions ([Raymo and Wang 2022](#)). Second, because research documents notable inequalities in the predictability of mortality for different social groups, the incorporation of subjective measures may improve predictive accuracy and thereby contribute to efforts to narrow mortality and longevity inequalities (e.g., by race).

3 Outcomes

Table 1: Outcome Descriptions

Outcome	Definition	Encoding
Longevity	Death age in years	Numeric (continuous)
Survive to 75	Whether the individual lives past 75 year of age	Numeric (binary)

4 Data

This project will use individual-level longitudinal from the Health and Retirement Study (HRS) to predict mortality. The HRS is a population-representative sample of 20,000 adults 55 years and older that contains information on income, wealth, health, education, and subjective expectations. The HRS also tracks mortality outcomes for the duration of individuals' participation in the study, making it an ideal dataset to use for predicting mortality because of its granularity, covariate richness, and sample size. The repeated-measures nature of these data also opens up additional possibilities for predictive performance because we observe individuals between 2 and 8 waves (4-16 years). Table 2 describes some key features of the sample. I anticipate that the N for this sample will be approximately 170,000 and the K will be between 82 – 100. This is before any binning or data transformations.

5 To Do

- Tables: Table comparing in and out-of sample GOF measures and percent improvement (over OLS) measures for continuous and binary predictors across models (OLS, Probit (for binary), Ridge, LASSO, Random Forest, XGBoost, and potentially Artificial Neural Network) between models with and without utility feature-subjective probabilities
- Figures: Calibration plots, AUC (for binary), variable importance plot, comparison between models with and without utility-feature-subjective probabilities

Table 2: Descriptive Statistics of Health and Retirement Study Sample

	N	Mean	Min	Max
Died	170411	0.32	0.00	1.00
Age	170411	64.51	50.00	79.00
Live to Age 75	170411	0.14	0.00	1.00
Black	170411	0.18	0.00	1.00
White	170411	0.75	0.00	1.00
Other	170411	0.07	0.00	1.00
Hispanic	170284	0.12	0.00	1.00
Female	170411	0.59	0.00	1.00
Subjective Life Expectancy (75+)	170411	64.40	0.00	100.00
Subjective Life Expectancy (80+)	170409	48.13	0.00	100.00
Less than HS	170411	0.25	0.00	1.00
HS	170411	0.47	0.00	1.00
Some College	170411	0.07	0.00	1.00
College	170411	0.13	0.00	1.00
Advanced Degree	170411	0.09	0.00	1.00
Wealth (Q1)	170411	0.19	0.00	1.00
Wealth (Q2)	170411	0.20	0.00	1.00
Wealth (Q3)	170411	0.20	0.00	1.00
Wealth (Q4)	170411	0.20	0.00	1.00
Wealth (Q5)	170411	0.20	0.00	1.00
Total Wealth (\$1,000s)	170411	592.09	-3482.96	155541.66
Total non-Housing Wealth (\$1,000s)	170411	392.11	-3482.96	154067.67
Total Stock Value (\$1,000s)	170411	82.07	-20.02	40826.75
N Children	170411	3.38	1.00	11.00
Mean Days of Contact with Children	156071	1283.21	1.00	8760.00
Mean Age of Children	159051	37.54	0.00	76.50
Sum of Functional Limitations	170411	0.30	0.00	6.00
Sum of Chronic Conditions	170411	1.87	0.00	8.00

All dollar values are inflation adjusted to 2022 Dollars per the CPI from the Bureau of Labor Statistics. Number of contact days with child is the number days over a two-year period. Number of observations varies by variable because certain values (e.g., subjective life expectancy) that are only observed for those over a certain age. Numbers of observations refer to person-observations over several HRS waves.

6 References

- Badolato, Luca, Ari Gabriel Decter-Frain, Nicolas Irons, Maria Laura Miranda, Erin Walk, Elnura Zhalieva, Monica Alexander, Ugofilippo Basellini, and Emilio Zagheni. 2023. *The Limits of Predicting Individual-Level Longevity*. Rostock.
- Breen, Casey, and Nathan Seltzer. 2023. “[Demographic Perspectives on Predicting Individual-Level Mortality](#).”
- Kose, Esra, UC Merced, Henry Manley, and Douglas L. Miller. n.d. “Backcasting Population Data for Children in the 1960s with Supervised Learning.”
- Raymo, James M., and Jia Wang. 2022. “Loneliness at Older Ages in the United States: Lonely Life Expectancy and the Role of Loneliness in Health Disparities.” *Demography* 59(3):921–47. doi: [10.1215/00703370-9937606](https://doi.org/10.1215/00703370-9937606).